CS 210

Program 3 Analysis

1. Briefly define a queue data structure.
What evidence did you find that a queue container was operating in the load balancer?
A queue is a first-in, first-out data structure. The first item added to the queue is the first item removed. I knew that the load balancer was using a queue because the requests were removed using front() and then removed with pop(). The output from the txt file shows that requests were processed in the same order they entered the load balancer.

2. Briefly define a stack data structure.
What evidence did you find that a stack container was operating in the load balancer?
A stack is a last-in, first-out data structure. The most recently added item is the first item removed. The load balancer used a stack because requests were accessed using top() then removed using pop(). The output from the txt file shows that newer requests were processed before older ones.

3. Briefly define a priority queue data structure.
What evidence did you find that a Max priority queue based on process time was operating in the load balancer?
A priority queue removes the item with the highest priority instead of using the order in which items were added. The MaxProcessTime priority queue gave the highest priority to requests with the longest process times. The output from the max process txt file shows that requests with larger process times were selected before requests with smaller process times.

What evidence did you find that a Min priority queue based on process time was operating in the load balancer?
The MinProcessTime priority queue gavve the highest priority to requests with the shortest process times. To compare, I did return r1.processTime > r2.processTime;. The output from the txt file showed that requests with smaller process times were selected before ones with larger process times.

What evidence did you find that a Max priority queue based on process ID was operating in the load balancer?
The MaxPID priority queue gave the highest priority to requests with the largest process IDs. To compare, I did return r1.pid < r2.pid;. The output from the txt file showed that requests with larger process IDs were selected before requests with smaller Ids.